MUHAMMAD KOMAIL KHAWAJA

AI ENGINEER

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Dedicated and enthusiastic Artificial Intelligence undergraduate with a strong foundation in Python, C++, SQL, machine learning, and data structures. Passionate about building AI-driven solutions and continuously expanding technical skills through academic projects and self-learning. A quick learner with strong problem-solving abilities, eager to contribute to innovative teams while gaining hands-on industry experience.

EDUCATION

Szabist University **2024–Present**
- Bachelor of Science in Artificial Intelligence

Punjab College of Science Blue Area Islamabad **2022-2024**
- Intermediate (ICS)

Adnan Hussain Shaheed Model College for boys **2020-2022**
- Matriculation (Science)

EXPERIENCE

Machine Learning Intern | CodeAlpha **2025-Present**
- Developed and optimized diverse machine learning and deep learning models using Python, Scikit-learn, and TensorFlow.
- Built a Credit Scoring Model using classification algorithms (Logistic Regression/Random Forest) to evaluate creditworthiness based on financial history, assessing performance via Precision, Recall, and ROC-AUC.
- Engineered an Emotion Recognition system using deep learning (CNN/LSTM), processing speech audio datasets (RAVDESS/TESS) by extracting Mel-Frequency Cepstral Coefficients (MFCCs).
- Designed a Handwritten Character Recognition system by training Convolutional Neural Networks (CNN) on image datasets (MNIST/EMNIST) for pattern recognition.
- Implemented predictive pipelines for Disease Prediction, utilizing classification techniques (SVM, Random Forest, XGBoost) on structured medical data from the UCI ML Repository.
- Maintained version-controlled public repositories on GitHub for all source code, demonstrating clear project documentation and model deployment readiness.

AI & Robotics Engineering Intern | Decode Lab **2026-Present**
- LLM Data Extraction & Prompt Engineering: Programmed large language models (LLMs) to perform zero-shot and few-shot text processing, utilizing strict token delimiters and structured in-context examples to achieve deterministic, valid JSON formatting without conversational overhead.
- Computer Vision Quality Inspection: Engineered an automated optical inspection framework using OpenCV to identify physical manufacturing anomalies in live image/video feeds, implementing custom Grayscale, Gaussian Blur, and thresholding preprocessing logic alongside contour bounding boxes.
- Autonomous Mobile Robot (AMR) Navigation: Developed navigation and obstacle-avoidance logic for a simulated wheeled robot using LiDAR sensor arrays, constructing a 2D Occupancy Grid mapping system and deploying A*or Dijkstra pathfinding algorithms for optimized routing.
- Industrial Automation & Control Logic: Programmed industrial control algorithms for a conveyor-belt sorting line using Ladder Logic and finite state machines, integrating physical sensor inputs with motor/pneumatic actuator outputs alongside strict emergency-stop safety interlocks.

SKILLS

Programming & Development
- C++
- Python
- SQL
- Object-Oriented Programming

Artificial Intelligence
- Machine Learning
- Deep Learning (Basics)
- Data Analysis
- Computer Vision (Basics)
- Natural Language Processing (Basics)

Databases & Networking
- Database Management Systems
- Computer Networks

Libraries & Frameworks
- Scikit-learn
- TensorFlow
- NumPy
- Pandas
- Matplotlib
- OpenCV
- Open-Source LLMs (Prompt Engineering)

Tools & Technologies
- Microsoft Office
- Git & GitHub
- Visual Studio Code
- Jupyter Notebook
- Google Colab

Soft Skills
- Problem Solving
- Communication
- Teamwork
- Time management
- Leadership

ACADEMIC PROJECTS

Software & Web Development

ZabDelivers – Food Delivery Web App | Database, Frontend, Backend
- Designed a relational database schema for users, multi-restaurant menus, and order logs; optimized queries using joins to handle concurrent traffic.
- Developed full-stack features including user authentication, real-time cart management, and dynamic vendor menu rendering.

ATM Machine Simulator | C++, OOP Principles
- Built an object-oriented desktop application featuring secure PIN validation, balance tracking, and deposit/withdrawal handling with robust error validation.

Hotel Management System | C++, File Handling
- Programmed a console application using OOP and file handling to ensure data persistence for guest records and transaction history.

AI & Machine Learning

Depression Detection System | Python, Scikit-Learn, Pandas, NLP
- Engineered an ML pipeline to clean and tokenize behavioral/textual datasets, training classification models to identify psychological indicators.

Car Resale Price Predictor | Python, Pandas, Matplotlib, Regression
- Conducted exploratory data analysis (EDA) and feature engineering to train a predictive regression model estimating vehicle values based on mileage and condition.

Energy Consumption Data Analysis | Data Structures, Algorithmic Analysis
- Conducted algorithmic analysis on large-scale energy datasets, applying graph/tree structures to map and identify peak-load inefficiencies.

Hardware & Networking

Enterprise Network Design & Simulation | Cisco Packet Tracer, VLSM, Routing Protocols

- Simulated a scalable corporate network architecture using VLSM subnetting, configuring inter-VLAN routing, DHCP pools, and Access Control Lists (ACLs).

RFID-Based Secure Door Lock | Assembly/C, Microcontroller, RFID Modules

- Programmed hardware interrupts and register-level logic to read scanner inputs and match UID keys against local storage to trigger electronic servos.

Smart Parking System | Digital Logic Design (DLD), IR Sensors, Logic Gates

- Built a hardware-driven circuit utilizing IR sensors and flip-flops to manage parking states and display available slots on a 7-segment display without a CPU.

Automated Street Light System | Arduino, LDR Sensor, Embedded C

- Developed an energy-efficient embedded system using an LDR sensor and ADC configurations to adjust LED brightness dynamically based on ambient light.

CERTIFICATIONS

Artificial Intelligence & Machine Learning

- Agents and Workflows | OpenAI Academy
  - Issued: July 2026 | Credential ID: 9s7vk2lmtc
- Applied AI Foundations | OpenAI Academy
  - Issued: July 2026 | Credential ID: tpxmber48i
- AI Foundations | OpenAI Academy
  - Issued: July 2026 | Credential ID: 5en1mthpyo
- AI Fundamentals with IBM SkillsBuild | Cisco Networking Academy & IBM
  - Issued: February 2026
- Introduction to Modern AI | Cisco Networking Academy
  - Issued: February 2026

Data Science & Development

- Intro to SQL | Kaggle
  - Issued: March 2026
- Introduction to Data Science | Cisco Networking Academy
  - Issued: February 2026
- Python Essentials 1 | Cisco Networking Academy (in partnership with OpenEDG Python Institute)
  - Issued: February 2026
- C++ Essentials 1 | Cisco Networking Academy (in partnership with C++ Institute)
  - Issued: February 2026

Computer Networking & Core Tools

- Exploring Networking with Cisco Packet Tracer | Cisco Networking Academy
  - Issued: March 2026
- Getting Started with Cisco Packet Tracer | Cisco Networking Academy
  - Issued: February 2026
- Getting Started with Microsoft Excel | Coursera (Offered by Freedom Learning Group)
  - Issued: February 2026 | Credential ID: COXGV4VP505T

LANGUAGES

- English
- Urdu